

# RISHIT RAMAN

+91-8092273454 | [ramanrishit448@gmail.com](mailto:ramanrishit448@gmail.com) | [linkedin.com/in/rishit-raman](https://linkedin.com/in/rishit-raman) | [github.com/ramanrishit448-RR](https://github.com/ramanrishit448-RR)

## Summary

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Software Engineering student focused on backend systems, AI applications, and modern web development — from a ground-up **C++17 vector database** (HNSW,  $O(\log N)$  search, RAG pipeline) to production platforms. Experienced across Next.js, tRPC, Prisma, LLM APIs (OpenAI, Gemini).

## Projects

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**VectorDB — Custom Vector Database Engine** [GitHub](#) Apr – May 2026  
(C++17, HNSW, KD-Tree, Ollama, RAG, REST API)

- Built a **private AI that reads and answers from your own documents** — no cloud, no API calls, runs entirely locally — powered by a **C++17 vector database** implementing **HNSW**, ( **$O(\log N)$** ), KD-Tree, and Brute Force with **3 distance metrics**, replicating Pinecone internals from scratch
- Implemented an end-to-end RAG pipeline with Ollama (768D embeddings + llama3.2) and 250-word overlapping chunking, articulating design choices to project stakeholders by **~60%** over flat search
- Exposed a **REST API with 10+ endpoints** and a React web UI with a live **PCA scatter plot**; HNSW delivered **~85% faster** queries than brute force at 1K+ vectors with **>95% recall** on benchmarks
- Benchmarked HNSW vs KD-Tree vs Brute Force — HNSW outperformed KD-Tree by **~3×** at 768D, with empirical data validating the curse of dimensionality

**Meet.AI — AI-Powered Video Conferencing Platform** [GitHub](#) Jan – Feb 2026  
(Next.js 15, React 19, TypeScript, OpenAI, Stream SDK, tRPC)

- Engineered a **custom AI Agent builder** where users define their own bot (role, behavior, instructions), and the moment a call starts, that exact agent joins as a **live voice participant** — not a chatbox, not a sidebar tool, but a real presence in the room via **OpenAI Realtime API**
- Integrated **OpenAI GPT-4** to auto-generate meeting transcripts and context-aware summaries, reducing post-meeting documentation time by **70%** (avg. 25 min/meeting)
- Optimized sub-millisecond transcript search using an **inverted-index** with timestamp-anchored video playback, accelerating meeting review workflows by **55%**
- Architected a type-safe backend with **tRPC** and **DrizzleORM** on Neon PostgreSQL; achieved **sub-100ms P95** query latency under concurrent load via connection pooling and query optimization

**Resonance — Open-Source Text-to-Speech SaaS** [GitHub](#) Dec 2025  
(Next.js, React 19, TypeScript, tRPC, Prisma, Cloudflare R2, Modal GPU, Clerk, Polar)

- Built a self-hostable, open-source **zero-shot voice cloning** SaaS (ElevenLabs alternative) on Next.js and React 19, supporting **20 voices across 12 categories and 5 locales**
- Deployed **Chatterbox TTS** on a serverless **NVIDIA A10G GPU via Modal**, enabling voice cloning from a 10-second sample with no fine-tuning — eliminating multi-hour training pipelines required by traditional cloning approaches
- Streamlined audio storage and retrieval on **Cloudflare R2** with type-safe **tRPC** routers and **Prisma ORM**, delivering waveform playback via WaveSurfer.js at **sub-200ms** proxy response times
- Designed a responsive, mobile-first UI with generation history, voice metadata persistence, and configurable TTS parameters (creativity, expression, pacing)

## Technical Skills

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**Languages:** JavaScript, TypeScript, C++, HTML/CSS  
**Frontend:** React.js, Next.js, Tailwind CSS, Redux  
**Backend:** Node.js, Express.js, tRPC, REST APIs, WebSockets  
**Databases:** Neonbase, MongoDB, Supabase, DrizzleORM, Prisma  
**AI/ML:** RAG, Vector Embeddings, Ollama  
**DevOps & Tools:** Git, GitHub, Cloudflare R2, Postman, Jest, npm

## Education

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**Dronacharya Group of Institutions, Greater Noida** 2023 – 2027  
*Bachelor of Technology in Information Technology*

**Jean Paul High School, Arrah** 2022 – 2023  
*Class 12 (Senior Secondary)*